

Assignment 6: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1 loading tidyverse, rvest, and here
library(tidyverse)
library(rvest)
library(here)
library(ggthemes)
library(cowplot)

#verifying my working directory
here()
```

```
## [1] "/home/guest/EDE_R_Lab_F26_NEW"
```

```
#Establishing my color palette
canva_pal("Birds and berries")
```

```
## function (n)
## {
##   n_values <- length(values)
##   if (n > n_values) {
##     cli::cli_warn("This manual palette can handle a maximum of {n_values} values. You have suppl
```

```
##     }
##     unname(values[seq_len(n)])
## }
## <bytecode: 0x5cfd575b7638>
## <environment: 0x5cfd575bcb70>
```

```
#Establishing my theme
my.theme <- theme_bw() +
  #making the title bold, 13 font, and centered
  theme(plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
    #making my axis titles bold and 10 font
    axis.title = element_text(face = "bold", size = 10),
    #positioning my legend on the right side of the plot
    legend.position = "right"
  )

#setting my theme
theme_set(my.theme)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
Durham_LWSP_Website <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024')
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - PWSID
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```

#3 creating the water system name variable
water_system_name <- Durham_LWSP_Website %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()

#creating the PWSID variable
PWSID <- Durham_LWSP_Website %>%
  html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
  html_text()

#creating the ownership variable
ownership <- Durham_LWSP_Website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()

#creating the maximum daily use for each month of 2024 variable
maximum_daily_use_monthly <- Durham_LWSP_Website %>%
  html_nodes("th~ td+ td") %>%
  html_text()

```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```

#4 creating a dataframe with the columns of the four variables I just scraped and a row for the months
Durham_2024_LWSP <- data.frame("water_system_name" = as.character(water_system_name),
                              "PWSID" = as.character(PWSID),
                              "ownership" = as.character(ownership),
                              "maximum_daily_use_monthly" = as.numeric(maximum_daily_use_monthly))

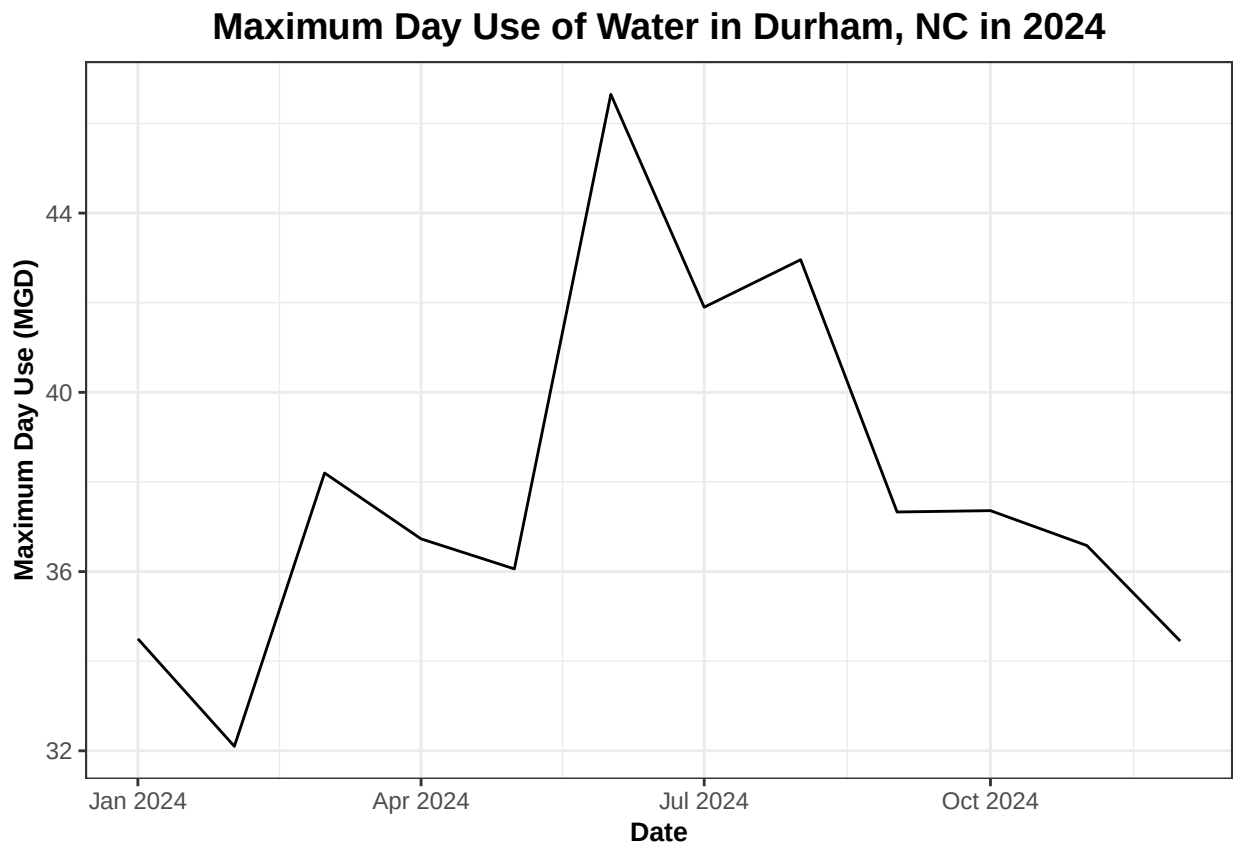
#adding a date column to the dataframe
Durham_2024_LWSP <- Durham_2024_LWSP %>%
  mutate(Date = my(paste(c("Jan", "May", "Sept", "Feb", "June", "Oct", "Mar", "Jul",
                          "Nov", "Apr", "Aug", "Dec"), 2024))) %>%
  mutate(Year = year(Date))

#viewing the dataframe and commenting it out after
#view(Durham_2024_LWSP)

#5 creating a line plot for the maxium daily use values for each month in 2024
ggplot(Durham_2024_LWSP, aes(x=Date, y=maximum_daily_use_monthly)) +
  geom_line() +

```

```
labs(title = "Maximum Day Use of Water in Durham, NC in 2024",
     x= "Date",
     y="Maximum Day Use (MGD)")
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6 creating a scraping function with PWSID and year
scrape.it.LWSP <- function(PWSID, Year){
  #retrieving the LWSP website contents
  NC_LWSP_Website <- read_html(paste0(
    'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=',PWSID,'&year=',Year))
  #setting the element variables
  PWSID_tag <- 'td tr:nth-child(1) td:nth-child(5)'
  Year_tag <- '.fancy-table:nth-child(64) tr:nth-child(1) th:nth-child(2)'
  water_system_name_tag <- 'div+ table tr:nth-child(1) td:nth-child(2)'
  ownership_tag <- 'div+ table tr:nth-child(2) td:nth-child(4)'
  maximum_daily_use_monthly_tag <- 'th~ td+ td'
```

```

#scraping the data items
PWSID_website <- NC_LWSP_Website %>%
  html_nodes(PWSID_tag) %>%
  html_text()

Year_website <- NC_LWSP_Website %>%
  html_nodes(Year_tag) %>%
  html_text()

#creating the dataframe
NC_LWSP <- data.frame("water_system_name" = as.character(water_system_name),
                      "ownership" = as.character(ownership),
                      "maximum_daily_use_monthly" =
                        as.numeric(maximum_daily_use_monthly)) %>%
  mutate(PWSID = !!PWSID,
         Year = !!Year,
         Date = my(paste(c("Jan", "May", "Sept", "Feb", "June", "Oct",
                           "Mar", "Jul", "Nov", "Apr", "Aug", "Dec"),
                          Year)))

#returns the dataframe as the function's output
return(NC_LWSP)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7 using my scrape.it function to extract the max daily use for Durham for each
#month in 2020 and return it in a dataframe
Durham_2020_LWSP <- scrape.it.LWSP('03-32-010', 2020)

#viewing the structure of my dataframe
glimpse(Durham_2020_LWSP)

```

```

## Rows: 12
## Columns: 6
## $ water_system_name      <chr> "Durham", "Durham", "Durham", "Durham", "Dur~
## $ ownership              <chr> "Municipality", "Municipality", "Municipalit~
## $ maximum_daily_use_monthly <dbl> 34.50, 36.06, 37.33, 32.10, 46.65, 37.36, 38~
## $ PWSID                  <chr> "03-32-010", "03-32-010", "03-32-010", "03-3~
## $ Year                    <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 20~
## $ Date                    <date> 2020-01-01, 2020-05-01, 2020-09-01, 2020-02~

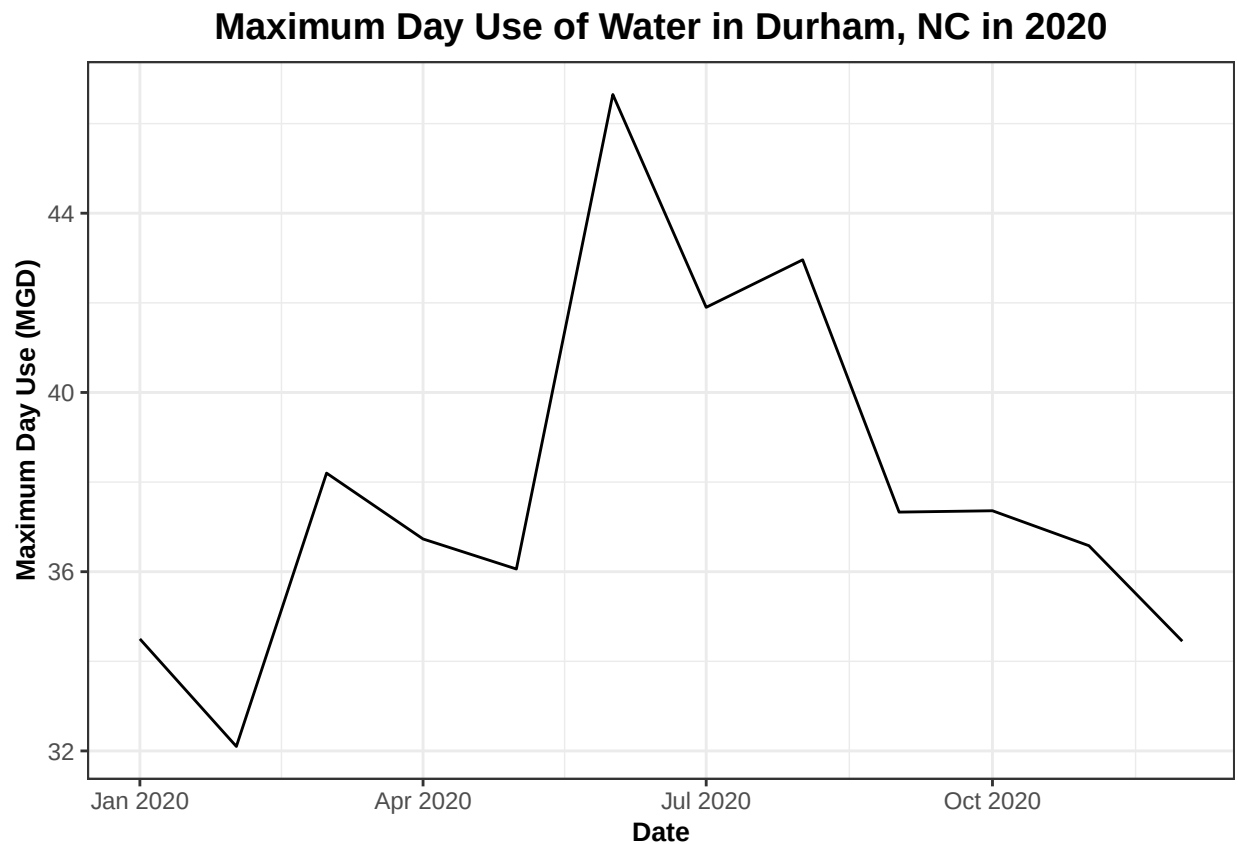
```

```

#plotting the date and maximum day use of water from the dataframe
Durham_2020_Plot <- Durham_2020_LWSP %>%
  ggplot(aes(x= Date, y= maximum_daily_use_monthly)) +
  geom_line() +
  labs(title = "Maximum Day Use of Water in Durham, NC in 2020",
       x= "Date",
       y="Maximum Day Use (MGD)")

```

```
#viewing the plot
Durham_2020_Plot
```



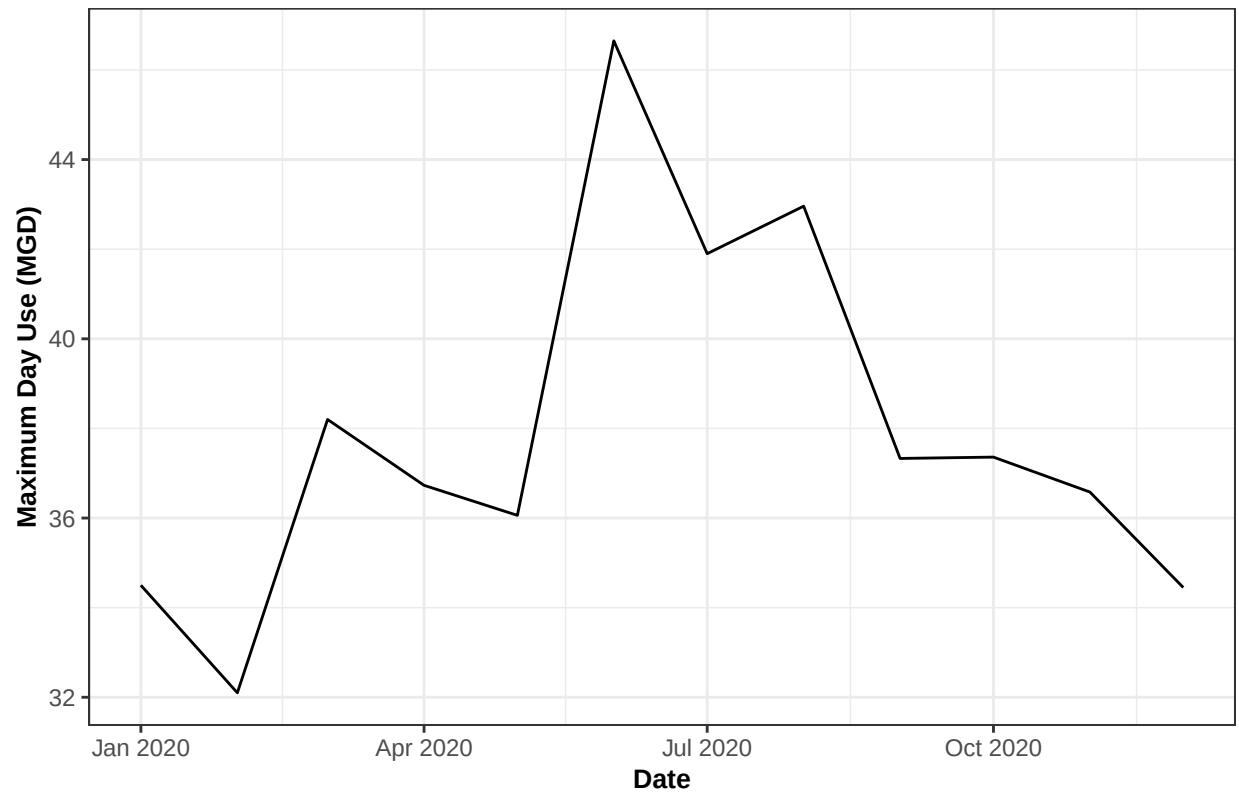
- Use the function above to extract data for Cary (PWSID = '03-32-010') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8 using my scrape.it function to extract the max daily use for Cary for each
#month in 2020 and return it in a dataframe
Cary_2020_LWSP <- scrape.it.LWSP('03-32-010', 2020)

#plotting the date and maximum day use of water from the dataframe
Cary_2020_Plot <- Cary_2020_LWSP %>%
  ggplot(aes(x= Date, y= maximum_daily_use_monthly)) +
  geom_line() +
  labs(title = "Maximum Day Use of Water in Cary, NC in 2020",
       x= "Date",
       y="Maximum Day Use (MGD)")

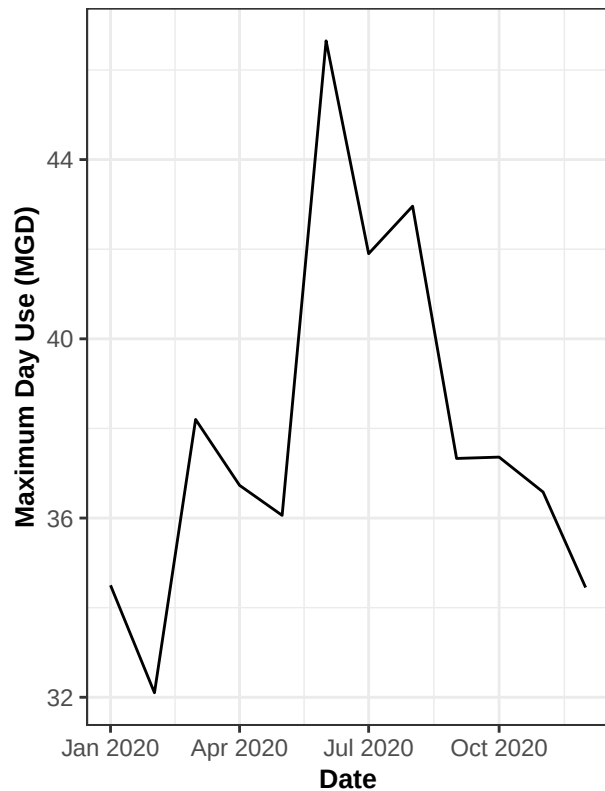
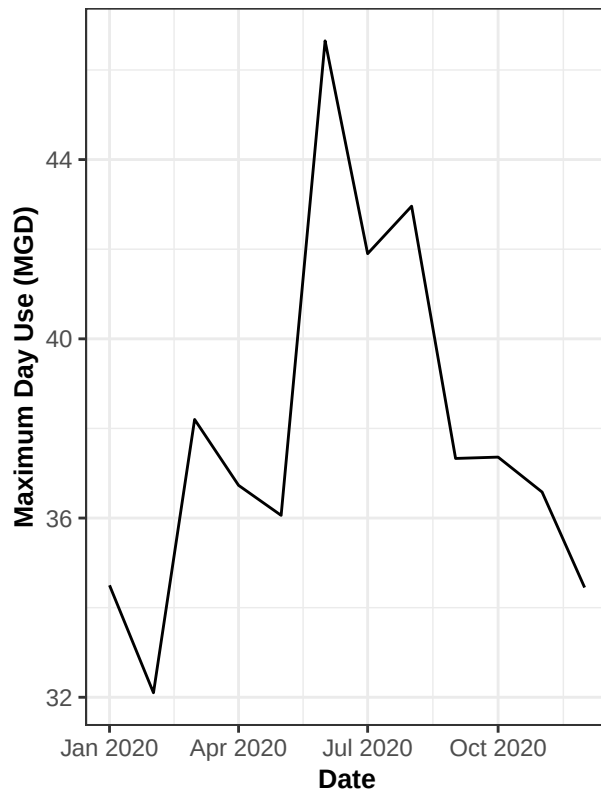
#viewing the plot
Cary_2020_Plot
```

Maximum Day Use of Water in Cary, NC in 2020



```
#using cowplot to combine my Durham 2020 and Cary 2020 maximum day use of water plots  
plot_grid(Durham_2020_Plot, Cary_2020_Plot)
```

Maximum Day Use of Water in Durham, NC



- Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9 creating a dataframe for Cary's maximum daily use by date for 2018-2023
Cary_2018_2023_LWSP <- map2('03-92-020', 2018:2023, scrape.it.LWSP)
```

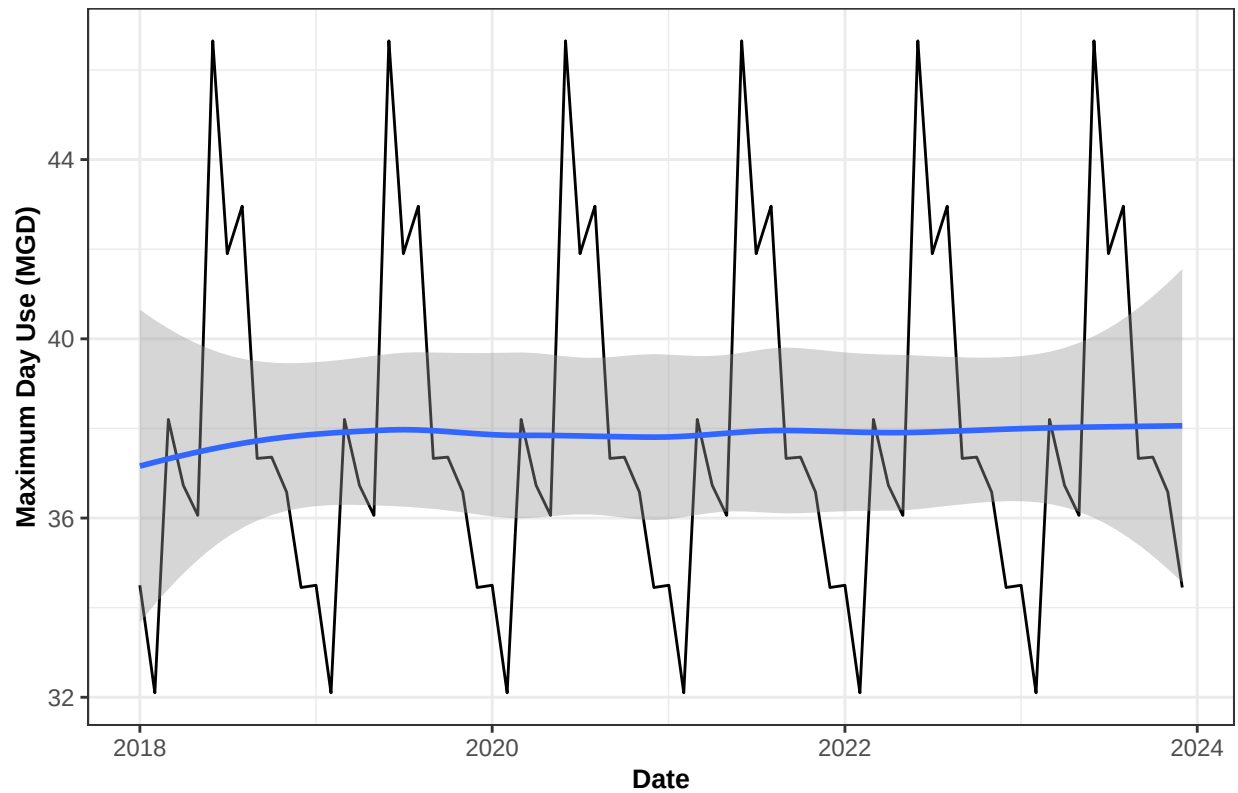
```
#combining these into one dataframe
Cary_2018_2023_LWSP_Combo <- bind_rows(Cary_2018_2023_LWSP)
```

```
#creating a plot from the combined dataframe with a smoothed line
Cary_2018_2023_Plot <- Cary_2018_2023_LWSP_Combo %>%
  ggplot(aes(x= Date, y= maximum_daily_use_monthly)) +
  geom_line() +
  geom_smooth(method="loess") +
  labs(title = "Maximum Day Use of Water in Cary, NC from 2018-2023",
       x= "Date",
       y="Maximum Day Use (MGD)")
```

```
#viewing the plot
Cary_2018_2023_Plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Maximum Day Use of Water in Cary, NC from 2018–2023



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: >